

Andrew D. Steen

Associate Professor
Department of Biology (Marine and Environmental Biology)
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Expertise and interests

Geomicrobiology of aquatic systems, life in slow ecosystems, microorganism-organic matter interactions, meta-analyses, data science for microbial ecology

Education and Professional Experience

August 2024-present	Associate Professor Department of Biology, Marine and Environmental Biology Section Department of Earth Sciences University of Southern California
2023-August 2024	Associate Professor Departments of Microbiology and Earth and Planetary Sciences University of Tennessee, Knoxville
2014-2023	Assistant Professor, University of Tennessee, Knoxville
2012-2014	Research Assistant Professor, University of Tennessee, Knoxville
2009-2011	Postdoctoral Associate, Center for Geomicrobiology, Aarhus University, Denmark
2009	Ph.D., Marine Sciences, University of North Carolina at Chapel Hill
2005	M.S., Marine Sciences, University of North Carolina at Chapel Hill
2000	Sc.B. with honors in Chemistry, Brown University

Federal Funding

2023-2027	Steen, AD. "CAREER: Do extraordinarily long enzyme lifetimes drive microbial community assembly in deep subsurface sediments?" NSF Biological Oceanography 2145434 (renewed as 2608079). <i>\$1,200,987.</i>
2023-2025	Learman, D, and AD Steen. "Collaborative Research: ANT LIA: Connecting metagenome potential to microbial function: Investigating microbial degradation of complex organic matter Antarctic benthic sediments". NSF Office of Polar Programs 2147046. <i>\$277,051 total, \$137,045 to Steen.</i>
2019-2022	Steen, AD, C Coleman-King, S Drumheller-Horton, K Rearden: "GP-IMPACT: ICE-AGE: Integrating Continuous Experiential Activities for Geoscience Education." NSF Improving Undergraduate STEM Education Grant 1911565. <i>\$351,411, \$81,887 to Steen.</i>
2019-2022	Lloyd, KG, J Cliff, R Hettich, TC Onstott, AD Steen, T Vishnivetskaya: "Using culture-independent methods to link active compound-specific carbon degradation to greenhouse gas production and recycling in natural populations of permafrost microbes". Department of Energy Biological and Environmental Research Grant DE-SC0020369. <i>\$3,320,280 total award, \$555,016 to Steen.</i>
2017	LaRowe, DA, AD Steen: "Proposal for a workshop: Microorganisms and Organic Carbon in the Marine Subsurface". NSF/Center for Dark Energy Biosphere Investigations (C-DEBI). <i>\$22,800.</i>

- 2016 McKay, LD, SP Horn; **AD Steen** as senior personnel: "GP-IMPACT: Engagement of Students and Faculty at Community Colleges to Enhance Recruitment to 4-Year Geoscience Programs". NSF-ICER. *\$415,988, \$24,952 to Steen.*
- 2014-2016 Lloyd, KG, **AD Steen**: "Organic matter degradation pathways and cell specific enzyme activities in uncultured microorganisms found in deep sediments of the Baltic Sea, IODP Leg 347". NSF Division of Ocean Sciences grant OCE-1431598. *\$362,095 total award, \$117,994 to Steen.*
- 2014-2017 A Buchan, **AD Steen**, A Stubbins, RG Spencer: "Collaborative Research: Marine priming effect – molecular mechanisms for the biomineralization of terrigenous dissolved organic matter in the ocean." NSF Division of Ocean Sciences grant OCE-1357242. *\$997,364 total award, \$162,383 to Steen.*
- 2013-2014 **Steen, AD**: "Novel Peptidases in Subsurface Sediments: Activities and Substrate Specificities": NSF/Center for Dark Energy Biosphere Investigations (C-DEBI). *\$49,970.*
- 2013-2014 KG Lloyd, **AD Steen**: NSF/Center for Dark Energy Biosphere Investigations (C-DEBI): "Comparative metagenomics of deep subsurface microbial communities". *\$148,422 total, \$32,460 to Steen.*

Competitive Funding from Other External Sources

- 2025 **Steen, AD**, K Michalska, M Radford, G Babnigg. "Structural characterization of potentially stable CAZymes from subsurface sediment bacteria and archaea. eBERlight proposal to Argonne National Laboratory Advanced Photon Source. In-kind grant of five beamtime shifts on beamline 21-ID-F. *Approx value: \$19,200.*
- 2025 **Contributor**: "Lake Victoria Drilling Project". Lead PI: Melissa Berke. International Continental Drilling Program (ICDP). ICDP contribution expected ~\$1.5 million.
- 2023 **Primary author**: "Proposal to fund a teacher to work with the ICE-AGE program", awarded by United Way of Greater Knoxville (UWKG) to East Tennessee Freedom School (ETFS). *Written with Denise Dean of ETFS. Because UWKG will not award funds to University of Tennessee, this grant to fund a teacher to participate in the ICE-AGE program was routed through ETFS. \$8000.*
- 2022 **Sole PI**: "Can the deep subsurface biosphere show us how to design ultra-stable enzymes?" NVIDIA Academic Hardware Grant Program. *One month exclusive access to A30 Tensor Core GPU w/ 4 TB storage. Approx. value \$8,400.*
- 2019 **Co-I**: "Diversity of functional genes in deeply branching uncultured microbes". XSEDE computational resource allocation. *1,600 node-hours at Texas Advanced Computing Center plus support from project specialist.*
- 2015 **Lead PI**: "Enzyme lability of sedimentary organic carbon". Deep Carbon Observatory. *\$22,688.*

Competitive Internal Funding

- 2025-2027 **Lead PI**: Wrigley Institute Faculty Innovation Award. "Capturing active enzymes to improve petroleum bioremediation." *\$50,000*
- 2022 **Lead PI**: Faculty Research Assistant Funding award. "Developing an HPLC-based assay for extracellular enzymes in marine sediments and soils." *\$1,500*
- 2022 **Lead PI**: "Does freezing increase CO₂ and CH₄ production rates in permafrost soils?". Co-written with Ph.D. student Jacob Perez. UT-K Faculty Senate Research Council Summer Graduate Research Assistantship. *\$4,500*
- 2018 **Lead PI**: "High performance computing approaches to identify novel genetic potential in uncultured microorganisms across the tree of life." Joint Institute for Computational Sciences Collaborative Grant. *In-kind grant of computational and technician resources. Approximate equivalent value of \$20,000.*

- 2018-2019 **Lead PI:** "Novel pathways for microcystin degradation in aquatic environments." UTK Multidisciplinary SEED grant. \$66,010.
- 2015 **Lead PI:** "Root mark analysis utilizing citizen science." UT Outreach Incentive Grant, \$2000.
- 2014 **Lead PI:** SARIF award: "Request for a High-Performance Liquid Chromatography system with fluorescence and photodiode-array detectors for environmental science". UTK SARIF award (internal) \$23,808.

Publications

[†]Undergraduate or high school students or teachers under my supervision

^{*}Graduate students or postdocs under my supervision

Peer-reviewed publications

58. Cramm, MA, Coskun ÖK, F Montemagno, M Selci, D Read, T Goodall, ^{*}B Green, S Mulay, K Sipes, A Abramov, C Heppell, J Boike, D Giovanelli, T Vishnevetskaya, R Hettich, **AD Steen**, KG Lloyd, W Orsi, A Jungblut, JA Bradley (2026). Seasonal thawing of high Arctic soils triggers selective microbial growth and predation. *mSystems* 0:e00738-25. DOI: 10.1128/msystems.00738-25
57. Abuah, C, K Sipes, J Buongiorno, **AD Steen**, R Liang, JA Bradley, D Giovannelli, A Abramov, S Peters, RJ Giannone, R Hettich, J Boike, T Vishnivetskaya, KG Lloyd (2026). Capacity of Arctic fjord sediments to degrade carbohydrates from permafrost active layer. *Microbiology Spectrum* 14: e00456-26. DOI: 10.1128/spectrum.00456-26
56. Sheik, CS, **AD Steen**, BK Reese, DT Wang, MR Osburn, KA Cantner, TL Kieft, FS Colwell, JR Spear, BM Toner (2026). Subsurface microbiology and the pressing societal need to support future exploration. *FEMS Microbiology Ecology* 102:1 fiae121. DOI: 10.1093/femsec/fiae121.
55. ^{*}Oduwale, I, M He, R Jha, A Hoarfrost, **AD Steen**, S Emrich (2025). Transformer-based prediction of microbial growth rates from genomic data. *IEEE Transactions on Computational Biology and Bioinformatics*. DOI: 10.1145/3765612.3767253
54. Lloyd, KG, **AD Steen** (2025). Defining ultra-slow-growing extremophilic microorganisms as aeonophiles (2025). *Nature Microbiology* 10: 1555-1557. DOI: 10.1038/s41564-025-02048-x
53. **Steen, AD**, J Buongiorno, K Rearden, C Coleman-King, S Drumheller-Horton, [†]E Phillips, [†]M Stokes, KG Lloyd (2025). Integrating authentic geoscience research into K-12 education via measurement of microbial extracellular enzymes in natural waters: a case report of two initiatives. In *GEOPATHS-Broadening the Voices in Geosciences: Pathways to innovative Pedagogy and Professional Development*, eds J MacDonald, R Clary, R Broadway, R Archer. GSA Books. DOI: 10.1130/2025.2564(02)
52. Howland, KE, HJ Nygaard, **AD Steen**, KM Halanych, AR Mahon, DR Learman (2025). Potential for microbial denitrification coupled with methanol oxidation found in abundant MAGs in Antarctic Peninsula sediments. *FEMS Microbiology Letters* fnaf050. DOI: 10.1093/femsle/fnaf050
51. Ruff, SE, IH de Angelis, M Mullis, JP Payet, C Magnabosco, KG Lloyd, CS Sheik, **AD Steen**, A Shipunova, A Morozov, BK Reese, JA Bradley, C Lemonnier, MO Schrenk, SB Joye, JA Huber, AJ Probst, HG Morrison, ML Sogin, J Ladau, F Colwell (2024). A global comparison of surface and subsurface microbiomes reveals large scale biodiversity gradients, and a marine-terrestrial divide. *Science Advances* 10(51): eadq0645. DOI: 10.1126/sciadv.adq0645
50. Coon, GR, LC Williams, A Matthews, R Diaz, RT Kevorkian, D LaRowe, AD Steen, L Lapham, KG Lloyd (2024). Control of hydrogen concentrations by microbial sulfate reduction in two contrasting anoxic coastal sediments. *Frontiers in Microbiology*. DOI: 10.3389/fmicb.2024.1455857
49. Sipes, K, J Buongiorno, **AD Steen**, AA Abramov, C Abuah, SL Peters, RJ Giannone, RL Hettich, J Boike, SL Garcia, TA Vishnevetskaya, KG Lloyd (2024). Depth-specific distribution of bacterial MAGs in permafrost active layer in Ny Ålesund, Svalbard (79°N). *Systematic and Applied Microbiology* 47(6):126544. DOI: 10.1016/j.syapm.2024.126544.
48. Rosengard, SZ, JMS Moura, RGM Spencer, C Johnson, A McNichol, **AD Steen**, V Galy (2024). Depth-partitioning of particulate organic carbon composition in the rising and falling stages of the

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Amazon River. *Geochemistry, Geophysics, Geosystems* 25(6):e2023GC011273.

DOI:10.1029/2023GC011273.

47. Shore, SFH, M Ptacek, **AD Steen**, EM Fozo (2024). A simple BLASTn-based approach generates novel insights into the regulation and biological function of type I toxin-antitoxins. *mSystems* e01204-23. DOI: 10.1128/msystems.01204-23.
46. Babjac, A, O Queen, SP Barhorst, K Kalhor, **AD Steen**, SJ Emrich (2023). Adapting protein language models for explainable fine-grained evolutionary pattern discovery. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. Dec 5: 2609-2616.
45. Mahmoudi, N, **AD Steen**, GP Halverson, KK Konhauser (2023). Biogeochemistry of Earth before exoenzymes. *Nature Geoscience* 16: 845–850. DOI: 10.1038/s41561-023-01266-4
44. Paul, R, TJ Rogers, KM Fullerton, M Selci, M Cascone, *MH Stokes, **AD Steen**, JM de Moor, A Chiodi, A Stefánsson, SA Halldórsson, CJ Ramirez, GL Jessen, PH Barry, A Cordone, D Giovannelli, KG Lloyd (2023). Complex organic matter degradation by secondary consumers in chemolithoautotrophy-based subsurface geothermal ecosystems. *PLoS ONE* 18(8): e0281277. DOI: 10.1371/journal.pone.0281277
43. Babjac, A, *TM Royalty, **AD Steen**, SJ Emrich (2022). A comparison of dimensionality reduction methods for large biological data. *The 13th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics*.
42. CJ Paradis, JI Miller, J-W Moon, SJ Spencer, LM Lui, JD Van Nostrand, D Ning, **AD Steen**, LD McKay, AP Arkin, J Zhou, EJ Alm, TC Hazen (2022). Sustained ability of a natural microbial community to remove nitrate from groundwater. *Groundwater*. 60(1) 99-111. DOI: 10.1111/gwat.13132
41. *Schmidt, JM, *TM Royalty, KG Lloyd, **AD Steen** (2021). Potential activities and long lifetimes of organic carbon degrading extracellular enzymes in deep subsurface sediments of the Baltic Sea. *Frontiers in Microbiology* 12:702015. DOI: 10.3389/fmicb.2021.702015
40. Fullerton KM, MO Schrenk, M Yücel, E Manini, M Basili, TJ Rogers, D Fattorini, M Di Carlo, G d'Errico, F Regoli, M Nakagawa, C Vetriani, F Smedile, C Ramírez, H Miller, SM Morrison, J Buongiorno, GL Jessen, **AD Steen**, M Martínez, JM de Moor, PH Barry, D Giovannelli, KG Lloyd (2021). Effect of tectonic processes on biosphere-geosphere feedbacks across a convergent margin. *Nature Geoscience*. DOI: 10.1038/s41561-021-00725-0
39. **Steen AD**, S Kusch, H Abdulla, N Cakić, S Coffinet, T Dittmar, J Fulton, V Galy, K-U Hinrichs, A Ingalls, B Koch, E Kujawinski, Z Liu, H Osterholz, D Rush, M Seidel, J Sepúlveda, SG Wakeham (2020). Analytical and computational advances, opportunities, and challenges in marine biogeochemistry in an era of “omics”. *Frontiers in Marine Science*. 7:718 DOI: 10.3389/fmars.2020.00718
38. Murray, AE, J Freudenstein, S Gribaldo, R Hatzepichler, P Hugenholtz, P Kämpfer, KT Konstantinidis, CE Lane, RT Papke, DH Parks, R Rosselló-Móra, MB Stott, IC Sutcliffe, JC Thrash, SN Venter, WB Whitman, SG Acinas, RI Amann, K Anantharaman, J Armengaud, BJ Baker, RA Barco, HB Bode, ES Boyd, CL Brady, P Carini, PSG Chain, DR Colman, KM DeAngelis, MA de los Rios, P Estrada-de los Santos, CA Dunlap, JA Eisen, D Emerson, TJG Ettema, D Eveillard, PR Girguis, U Hentschel, JT Hollibaugh, LA Hug, WP Inskeep, EP Ivanova, H-P Klenk, W-J Li, KG Lloyd, FE Löffler, T Makhalanyane, DP Moser, T Nunoura, M Palmer, V Parro, C Pedrós-Alió, AJ Probst, THM Smits, **AD Steen**, ET Steenkamp, A Spang, FJ Stewart, JM Tiedje, P Vandamme, M Wagner, FP Wang, BP Hedlund, A-L Reysenbach (2020). Roadmap for naming uncultivated microorganisms. *Nature Microbiology*. DOI: 10.1038/s41564-020-0733-x
37. LaRowe, DA, S Arndt, J Bradley, E Estes, A Hoarfrost, SM Lang, KG Lloyd, N Mahmoudi, W Orsi, SS Walter, **AD Steen**, R Zhao (2020). The fate of organic carbon in marine sediments - new insights from recent data and analysis. *Earth Science Reviews*. 204: 103146. DOI: 10.1016/j.earscirev.2020.103146

36. Mahmoudi, N, [†]SM Hagen, TC Hazen, **AD Steen** (2020). Patterns in extracellular enzyme activity and microbial diversity in deep-sea Mediterranean sediments. *Deep-Sea Research Part I*. DOI: 10.1016/j.dsr.2020.103231
35. Lang, SQ, MR Osbourn, **AD Steen** (2019). Carbon in the Deep Biosphere: Forms, Fates, and Biogeochemical Cycling. In *Whole Earth Carbon: Past and Present*, eds. B Orcutt, I Daniel, R Dasgupta. Cambridge: Cambridge University Press. Selected by CHOICE360, a publishing unit of the American Libraries Association, as an Outstanding Academic Title of 2020.
34. ^{*}Royalty, TM, **AD Steen** (2019). Theoretical and simulation-based investigation of the relationship between sequencing effort, microbial community richness, and diversity in binning metagenome-assembled genomes. *mSystems*. 4(5):e00384-19. DOI: 10.1128/mSystems.00384-19
33. ^{*}Royalty, TM, **AD Steen** (2019). Quantitatively partitioning microbial genomic traits among taxonomic ranks across the microbial tree of life. *mSphere*. 4(4):e00446-19
32. Senter, JK, ^{*}TM Royalty, **AD Steen**, A Sadovnik (2019). Unaligned Sequence Similarity Search Using Deep Learning. 2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), San Diego, CA: 1892-1899. DOI: 10.1109/BIBM47256.2019.8983072.
31. Ziervogel, K, SB Joye, S Kleindienst, SY Makin, U Passow, **AD Steen**, C Arnosti (2019). Polysaccharide hydrolysis in the presence of oil and dispersants: insights into potential degradation pathways of exopolymeric substances (EPS) from oil-degrading bacteria. *Elementa: Science of the Anthropocene*. 7(1), p.31. DOI: 10.1525/elementa.371
30. **Steen, AD**, ^{*}RT Kevorkian, JT Bird, N Dombrowski, BJ Baker, [†]SM Hagen, [†]KH Mulligan, ^{*}JM Schmidt, [†]AT Webber, ^{*}TM Royalty, MJ Alperin (2019). Kinetics and identities of extracellular peptidases in subsurface sediments of the White Oak River Estuary, NC. *Applied and Environmental Microbiology*. 85(19):e00102-19. DOI: 10.1128/AEM.00102-19
29. **Steen, AD**, A Crits-Christoph, P Carini, KM DeAngelis, N Fierer, KG Lloyd, JC Thrash (2019). High proportions of bacteria and archaea across most biomes remain uncultured. *ISME Journal* 13:3126-3130. doi: 10.1038/s41396-019-0484-y. **Selected as one of 20 "Best of the ISME Journal 2019" papers**
28. Sala, MM, J Piontek, S Endres, AM Romaní, S Dyhrman, **AD Steen** (2019). Extracellular enzymes in aquatic environments: exploring the link between genomic potential and biogeochemical processes. *Frontiers in Microbiology*. 10:1463. DOI: 10.3389/fmicb.2019.01463
27. Bird, JT, ED Tague, L Zinke, ^{*}JM Schmidt, **AD Steen**, B Reese, IPG Marshall, G Webster, A Weightmann, HF Castro, SR Campagna, KG Lloyd (2019). Uncultured microbial phyla suggest mechanisms for multi-thousand-year subsistence in Baltic Sea sediments. *mBio* 10 (2) e02376-18; DOI: 10.1128/mBio.02376-18
26. ^{*}Quigley, LMN¹, A Edwards, **AD Steen**, A Buchan (2019). Characterization of the interactive effects of labile and recalcitrant organic matter on microbial growth and metabolism. *Frontiers in Marine Science*. 10:493. DOI: 10.3389/fmicb.2019.00493
25. Lloyd, KG, **AD Steen**, J Ladau, J Yin, L Crosby (2018). Phylogenetically novel uncultured microbial cells dominate Earth microbiomes. *mSystems* 3(5) e-00055-18; DOI: 10.1128/mSystems.00055-18
24. Matzek, L, M Tipton, A Farmer, **AD Steen**, K Carter (2018). Understanding electrochemically activated persulfate and its application to ciprofloxacin abatement. *Environmental Science and Technology* 52(10): 5875-5883
23. ^{*}Mullen, L, [†]Malcolm X Shabazz Aquatic Geochemistry Team, [†]K Boerrigter, [†]N Ferriero, [†]J Rosalsky, ^{*}AvB Barrett, [†]PJ Murray, **AD Steen** (2018). Potential activities of freshwater exo- and endo-acting extracellular peptidases in East Tennessee and the Pocono Mountains. *Frontiers in Microbiology*. 9:368. DOI: 10.3389/fmicb.2018.00368

¹ Co-advised with Dr. Alison Buchan as primary advisor.

22. Mahmoudi, N, SR Beaupré, **AD Steen**, and A Pearson (2017). Sequential bioavailability of sedimentary organic matter to heterotrophic bacteria. *Environmental Microbiology* 19(7): 2629-2644. DOI: 10.1111/1462-2920.13745
21. **Steen, AD**, ^{*}LNM Quigley and A Buchan (2016). Evidence for the priming effect in a planktonic estuarine microbial community. *Frontiers in Marine Science* 3:6. DOI: 10.3389/fmars.2016.00006
20. Michalska, K, **AD Steen**, G Chhor, M Endres, [†]AT Webber, J Bird, KG Lloyd, A Joachimiak (2015). New aminopeptidase from "Microbial Dark Matter" Archaeon. *FASEB Journal*. DOI: 10.1096/fj.15-272906.
19. **Steen, AD**, [†]JP Vazin, [†]SM Hagen, [†]KH Mulligan, and SW Wilhelm (2015). Substrate specificity of aquatic extracellular peptidases assessed by competitive inhibition assays using synthetic substrates. *Aquatic Microbial Ecology* 75(3) 271-281. DOI: 10.3354/ame01755
18. Cardman, Z, C Arnosti, A Durbin, K Ziervogel, C Cox, **AD Steen**, A Teske (2014). Verrucomicrobia Are Candidates for Polysaccharide-Degrading Bacterioplankton in an Arctic Fjord of Svalbard. *Applied and Environmental Microbiology* 80(12): 3749-3756.
17. **Steen, AD** and C Arnosti (2014). Picky, hungry eaters in the cold: Persistent substrate selectivity among polar pelagic microbial communities. *Frontiers in Microbiology* DOI: 10.3389/fmicb.2014.00527
16. Arnosti, C, C Bell, D Moorhead, RL Sinsabaugh, **AD Steen**, M Stromberger, M Wallenstein, MN Weintraub (2014). Extracellular enzymes in terrestrial, freshwater, and marine environments: New perspectives on system variability and common research needs. *Biogeochemistry* 117:5-21. DOI: 10.1007/s10533-013-9906-5
15. Lloyd, K, M K May, RT Kevorkian, and **AD Steen** (2013). Meta-analysis of quantification methods shows archaea and bacteria to be similarly abundant in the seafloor. *Applied and Environmental Microbiology* 79(24): 7790-7799. DOI: 10.1128/AEM.02090-13. **Selected by editors as an "Article of Significant Interest"**
14. Arnosti, C, and **AD Steen** (2013). Patterns of extracellular enzyme activities and microbial metabolism in an Arctic fjord of Svalbard and in the northern Gulf of Mexico: Contrasts in carbon processing by pelagic microbial communities. *Frontiers in Microbiology* 4:318 DOI:10.3389/fmicb.2013.00318
13. **Steen, AD**, BB Jørgensen, and B Aa Lomstein (2013). Abiotic racemization kinetics of amino acids in marine sediments. *PLOS ONE* 8(8) e71648. DOI:10.1371/journal.pone.0071648
12. Lloyd, KG, L, Schreiber, DG Petersen, K Kjeldsen, MA Lever, **AD Steen**, R Stepanauskas, M Richter, S Kleindienst, S Lenk, A Schramm, BB Jørgensen (2013). Predominant Archaea in marine sediments degrade detrital proteins. *Nature* 496: 215-218. DOI: 10.1038/nature12033
11. **Steen, AD**, and C Arnosti (2013). Extracellular peptidase and carbohydrate hydrolase activities in an Arctic fjord (Smeerenburgfjord, Svalbard). *Aquatic Microbial Ecology* 69: 93-99.
10. **Steen, AD**, K Ziervogel, S Ghobrial, and C Arnosti (2012). Functional variation among polysaccharide hydrolyzing microbial communities in the Gulf of Mexico. *Marine Chemistry* 138-139: 13-20
9. **Steen, AD** and K Ziervogel (2012). Comment on the review by German et al (2011) "Optimization of hydrolytic and oxidative enzyme methods for ecosystem studies" [Soil Biology & Biochemistry 43: 1387-1397]. *Soil Biology and Biochemistry* 48: 196-197
8. Arnosti, C, **AD Steen**, K Ziervogel, S Ghobrial, and WH Jeffrey (2011). Latitudinal gradients in degradation of marine dissolved organic carbon. *PLOS ONE* 6(12): e28900. DOI:10.1371/journal.pone.0028900
7. **Steen, AD**, and C Arnosti (2011). Long lifetimes of β -glucosidase, leucine aminopeptidase, and phosphatase in Arctic seawater. *Marine Chemistry* 123(1-4) 127-132

6. **Steen, AD**, K Ziervogel, and C Arnosti (2010). Comparison of multivariate microbial datasets with the Shannon index: An example using enzyme activity from diverse marine environments. *Organic Geochemistry* 41(9): 1019-1021
5. Ziervogel, K, **AD Steen**, and C Arnosti (2010). Changes in the spectrum of extracellular enzyme activities in seawater following aggregate formation. *Biogeosciences* 7: 1007-1015
4. Robador, A, V Bruchert, **AD Steen**, and C Arnosti (2010). Temperature induced decoupling of enzymatic hydrolysis and carbon remineralization in long-term incubations of Arctic and temperate sediments. *Geochimica et Cosmochimica Acta* 74(8): 2316-2326
3. **Steen, AD**, P Gururaj, J Ma, NV Blough, C Arnosti (2008). Fluorescence anisotropy as a means to determine extracellular polysaccharide hydrolase activity in environmental samples. *Analytical Biochemistry* 383(2): 340-342
2. **Steen, AD**, LJ Hamdan, and C Arnosti (2008). Dynamics of dissolved carbohydrates in the Chesapeake Bay: Insights from enzyme activities, concentrations, and microbial metabolism. *Limnology and Oceanography* 53(3): 936-947.
1. **Steen, AD**, C Arnosti, L Ness, and NV Blough (2006). Electron paramagnetic resonance spectroscopy as a novel approach to measure macromolecule-surface interactions and activities of extracellular enzymes. *Marine Chemistry* 101: 266-276

Software product

*Cook, CS and **AD Steen**. ezmek. An R package to easily analyze enzyme kinetics data from fluorogenic substrate proxies. Available on CRAN: <https://cran.r-project.org/web/packages/ezmek/index.html>

Submitted manuscripts and preprints

Howland, K, A Suchy, J Shablin, H Nygaard, M Henson, **AD Steen**, K Halanych, G Colores, P Kourtev, T Hoellein, A Mahon, DR Learman (submitted). Abundant MAGs Drive Sediment Microbial Microplastic Degradation Potential Spanning the Antarctic Continental Shelf.

Sanger, A, JS Boden, **AD Steen**, EP Mueller, KO Konhauser, E Stüeken, RE Anderson, N Mahmoudi (submitted). Tracing the evolution of microbial alkaline phosphatases and their role in phosphorus recycling through time.

Sanger, A, **AD Steen**, JS Boden, S Cellier-Goetghebeur, M Bayder, EP Mueller, GP Halverson, NB Cowan, RE Anderson, JN Pelletier, E Stüeken, M Fakhraee, KO Konhauser, N Mahmoudi (2025). Microbial extracellular enzymes catalyzed the transition to an oxygenated Earth. bioRxiv. <https://www.biorxiv.org/content/10.1101/2025.09.11.675588v1>

*Oduwole, I, A Babjac, *TM Royalty, †M Hibbs, KG Lloyd, SJ Emrich, **AD Steen** (2025). Functional genomic signatures predict microbial culturability across the tree of life. bioRxiv. <https://doi.org/10.1101/2025.08.18.670795>

*Oduwole, I, T Vishnevetskaya, **AD Steen** (2025). Thermophilic traits correlate with slow growth in permafrost soils. bioRxiv. <https://www.biorxiv.org/content/10.1101/2025.07.23.666387v1>.

*Perez, JT, F Abuah, MA Cramm, SA Mulay, M Selci, K Sipes, AA Abramov, JA Bradley, J Boike, D Giovannelli, TA Vishnevetskaya, E Herndon, KG Lloyd, DJ Keffer, **AD Steen** (2025). Surface Charge Distributions of Predicted Soil Extracellular Enzymes Influence Mineral Adsorption and Enzyme Flexibility. Available at SSRN: <https://ssrn.com/abstract=5257736> or <http://dx.doi.org/10.2139/ssrn.5257736>

*Royalty, TM, **AD Steen**. Functional redundancy in ocean microbiomes controls trait stability. bioRxiv. <https://doi.org/10.1101/2021.06.18.448980>.

*Royalty, TM, **AD Steen**. A quantitative measure of functional redundancy in microbial ecosystems. bioRxiv. <https://doi.org/10.1101/2020.04.22.054593>.

White papers

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2. Toner, B, J Spear, B Kiel Reese, C Sheik, **AD Steen** (2024). Geomicrobiology Science Planning for Continental Drilling and Coring 2024.
<https://sites.google.com/umn.edu/csdsienceplanning/home/geomicrobiology-executive-summary>
1. Hand, K., [67 authors], **AD Steen**, [7 authors], Anaïs Roussel. On the Past, Present, and Future Role of Biology in NASA's Exploration of our Solar System. Whitepaper #229, Planetary Science and Astrobiology Decadal Survey 2023-2032. Bulletin of the American Astronomical Society. 53(4) Planetary Astrobiology Decadal Survey Whitepapers. DOI: 10.3847/25c2cf.7272e5bb

As a member of a group author

Beck, CC, M Berke, CS Feibel, V Foester, L Olaka, HM Roberts, CA Scholz, K Cantner, A NOren, GM Kiptoo, J Muirhead, and the **Deep Drilling in Turkana Basin (DDTB) Project Team** (submitted). ICDP workshop on the Deep Drilling in the Turkana Basin Project: Exploring the link between environmental factors and hominin evolution over the past 4 Ma. Scientific Drilling.

Publications from lab members on which I am not an author

I encourage students in my lab to make scientific contributions wherever they are able. Consequently, my lab members occasionally earn authorship on papers on which I am not an author. The following is a list of additional publications on which students in my lab have earned authorship, on which I am not an author.

6. Kelleher, JM, [18 co-authors], *I Oduwole, [12 co-authors], EA Eloë-Fadrosh (2025). A cost and community perspective on the barriers to microbiome data reuse. *Frontiers in Bioinformatics* 5:1585717. DOI: 10.3389/fbinf.2025.1585717
5. Li, Y, K Ash, DC Joyner, DE Williams, I Alamilla, P McKay, C Iler, *BM Green, F Kara-Murdoch, C Swift, TC Hazen (2023). Decay of enveloped SARS-CoV-2 and non-enveloped PMMoV RNA in raw sewage from university dormitories. *Frontiers in Microbiology* 14:1144026. DOI: 10.3389/fmicb.2023.1144026
4. Ash, KT, Y Li, I Alamilla, DC Joyner, P McKay, C Iler, *BM Green, F Kara-Murdoch, C Swift, TC Hazen (2023). SARS-CoV-2 Raw Wastewater Surveillance from Student Residences on an Urban University Campus. *Frontiers in Microbiology*. 14:1101205. DOI: 10.3389/fmicb.2023.1101205.
3. Ash, KT, I Almillia, Y Li, DC Joyner, DE Williams, PJ McKay, *BM Green, C Iler, SE DeBlander, FK Murdoch, C Swift, TC Hazen (2021). Coding-complete genome sequence of a SARS-CoV-2 variant obtained from raw sewage on University of Tennessee-Knoxville campus. *Microbial Resource Announcements* 10(47):e01049-21. DOI: 10.1128/MRA.01049-21.
2. Herbert, LC, Q Zhu, AB Michaud, K Laufer-Meiser, CK Jones, N Riedinger, †ZS Stooksbury, RC Aller, BB Jørgensen, LM Wehrmann (2021). Benthic iron flux influenced by climate-sensitive interplay between organic carbon availability and sedimentation rate in Arctic fjords. *Limnology and Oceanography* 66(9). DOI: 10.1002/lno.11885.
1. Gladstone NS, *TM Royalty, EM Carter, ML Niemiller (2020). *Eurycea lucifuga* (Cave salamander). Diet. (Field note). *Herpetological Review* 51(2).

Invited seminars and talks

- Steen, AD** (2025). Keynote: Deep learning, machine learning and AI in microbial ecology: A skeptic's guide. Society for Aquatic Microbial Ecology (SAME) 18th meeting (Barcelona, Spain).
- Steen, AD** (2024). Insights into the subsurface lifestyle from microbial extracellular enzymes. Kent State University Department of Earth Sciences weekly seminar (Kent, OH).
- Steen, AD** (2023). Insights into the subsurface lifestyle from microbial extracellular enzymes. University of Minnesota Biotechnology Institute and University of Minnesota-Duluth Department of Biology weekly seminars (Minneapolis and Duluth, MN).
- Steen, AD** (2023). What extracellular enzymes tell us about the state of the ocean. Skidaway Institute of Oceanography weekly seminar (Skidaway Island, GA).

- Steen, AD** (2022). Using machine learning to understand the ecology of subsurface microbes. UT-Knoxville -omics Symposium (Online).
- Steen, AD** (2022). Life in the deep marine subsurface. Department of Geology and Geological Engineering weekly seminar, University of Mississippi (Oxford, MS).
- Steen, AD** (2022). 'It's evolution, Jim, but not as we know it': Adaptation and filtering as drivers of microbial evolution in deep marine sediments. High Pressure Structural Biology and Extreme Biophysics (Ithaca, NY).
- Steen, AD**, ^{*}TM Royalty, ^{*}Schmidt, JM, KG Lloyd (2022). Do deep marine sediments host unusually stable enzymes? 66th Annual Biophysical Society meeting (San Francisco, CA).
- Steen, AD**, J Senter, ^{*}TM Royalty, A Sadovnik (2019). Linking metagenomes to organic geochemistry: How to deal with hypothetical and poorly-annotated genes. Hanse-Wissenschaftskolleg Workshop on Marine Organic Biogeochemistry (Delmenhorst, Germany).
- Steen, AD** (2018). Work-life balance is essential and also it can't possibly exist. C-DEBI Professional Development Webinar (online). Archived at <https://www.darkenergybiosphere.org/outputs-resources/professional-development-webinar-series/>
- Steen, AD** (2017). Tracing the carbon cycle through biomolecules. DCO Global Carbon Cycles Workshop (Moscow).
- Steen, AD** (2017). How heterotrophic microorganisms access sedimentary organic matter: A few known knowns, and plenty of known unknowns. American Chemical Society Spring Meeting.
- Steen, AD** (2016). Heterotrophic organisms in deep marine sediments: or, What to do when you're hungry but you're 100,000 years late for dinner. ORION (Oak Ridge amateur astronomy and science club) monthly meeting (Oak Ridge TN)
- Steen, AD** (2016). Bioavailability of organic carbon in deep sediments. School of Marine and Atmospheric Sciences weekly seminar, Stony Brook University (Stony Brook, NY).
- Steen, AD** (2016). Bioavailability of organic carbon in deep sediments. Marine Science Program weekly seminar, University of South Carolina (Columbia, SC)
- Steen, AD** (2016). How to make recalcitrant organic matter bioavailable: or, What silverware do you use when you're 10²-10⁶ years late for dinner? C-DEBI Annual Meeting (Marina, CA).
- Steen, AD** (2015). The revolution in marine dissolved organic matter chemistry and what it means for geomicrobiology. USC/Agouon International Geobiology Course. (Catalina Island, CA).
- Steen, AD** (2015). Using peptidase biochemistry to better understand protein degradation in aquatic environments. Massachusetts Institute of Technology Chemical Oceanography-Geobiology weekly seminar; Woods Hole Oceanographic Institution Chemical Oceanography weekly seminar.
- Steen, AD** (2015). Extracellular enzymes in the environment: What do we really know? China-US Workshop on Biogeochemistry and Climate Change. Institute of Applied Ecology – Chinese Academy of Sciences (Shenyang, China); Shenyang Agricultural University (Shenyang, China) and Inner Mongolia University (Hohhot, China).
- Steen, AD** (2015). Beyond V_{max} and K_m : How details of enzyme function influence geochemical cycles. AGU Fall Meeting (San Francisco, CA).
- Steen, AD** (2015). Microbial carbon oxidation in marine sediments: A biochemical perspective. Second Deep Carbon Observatory Early Career Scientist Workshop (Ponta Delgada, Azores).
- Steen, AD** (2011). Dynamics of marine extracellular enzymes: What we can (and can't) learn from using nonstandard substrates. Uppsala University, Sweden.
- Steen, AD** (2011). Biological influence on dissolved organic matter reactivity. Skidaway Institute of Oceanography, Skidaway Island, GA.
- Steen, AD** (2011). Biological influence on dissolved organic matter reactivity. Technical University of Denmark (DTU), Copenhagen, Denmark.
- Steen, AD** and C. Arnosti (2004). Measurement of extracellular enzyme activities in aquatic environments. U.S. Naval Research Laboratory, Washington, DC.

Conference abstracts, talks, and posters

[†]Undergraduate or high school students or teachers under my supervision

*Graduate students or postdocs under my supervision

My ocean sciences talk

Jake's ocean sciences talk

Allie's ocean sciences talk?

Vishnevetskaya, TA, JT Perez, I Oduwole, CY Abuah, KM Sipes, AA Abramov, AD Steen, KG Lloyd (2026). Methane emitted from ground surface and borehole depths: Ny-Ålesund, Svalbard. 10th International Conference on Polar and Alpine Microbiology (Copenhagen, Denmark).

Lloyd, TG, CY Abuah, K Sipes, J Buongiorno, AD Steen, JA Bradley, D Giovanelli, A Abramov, SL Peters, RJ Giannone, RL Hettich, J Boike, TA Vishnevetskaya (2024). Organic carbon remineralization in some surprising places in Arctic soils. The Sixth Meeting of Bergey's International Society for Microbial Systematics (BISMIS; Qingdao, China).

*Oduwole, I, TA Vishnevetskaya, **AD Steen** (2024). Is thermophilicity an adaptation to slow growth in cold, subsurface microbes? ISME19 (Cape Town, South Africa).

Liang, R, KG Lloyd, SL Peters, TA Vishnevetskaya, K Sipes, **AD Steen**, A Abramov, J Boike, RL Hettich, TC Onstott (2024). Dynamics of greenhouse gas fluxes and microbial communities associated with warming of subsurface permafrost cores. AGU Fall Meeting (Washington, DC).

*Matthews, A, *JT Perez, GR Coon, K Howland, K Nygaard, S Flaherty, DR Learman, **AD Steen** (2024). Geochemistry and geomicrobiology of the Eastern Antarctic continental shelf. 11th Scientific Committee on Antarctic Research (SCAR) 2024 Open Science Conference (Pucón, Chile).

*Oduwole, I, TA Vishnevetskaya, **AD Steen** (2024). Is thermophilicity an adaptation to slow growth in the cold subsurface? Southeastern Biogeochemistry Symposium (Atlanta, GA).

*Kalhor, K, *TM Royalty, **AD Steen** (2024). Assessment of enzyme stability in subsurface sediments by computational methods. Southeastern Biogeochemical Symposium (Atlanta, GA).

*Perez, JT, *B Green, *M Stokes, M Cramm, S Mulay, M Selci, K Sipes, A Abramov, JA Bradley, J Boike, D Giovanelli, TA Vishnevetskaya, KG Lloyd, DJ Keffer, **AD Steen** (2024). Extracellular enzyme flexibility during adsorption to mineral surfaces: A molecular dynamics study of metagenomically-derived Arctic enzymes. Southeastern Biogeochemistry Symposium (Atlanta, GA).

*Matthews, A, *JT Perez, GR Coon, K Howland, K Nygaard, S Flaherty, DR Learman, **AD Steen** (2024). The Eastern continental shelf of Antarctica: Investigating geochemistry and extracellular enzymes. Southeastern Biogeochemistry Symposium (Atlanta, GA).

Howland, K, H Nygaard, **AD Steen**, D Learman (2024). Metagenomic characterization of anaerobic hydrocarbon degradation in benthic marine sediment core from the Antarctic Peninsula. Goldschmidt conference (Chicago, IL).

Lloyd, KG, TA Vishnevetskaya, **AD Steen**, R Hettich, R Liang, J Cliff, TC Onstott, A Abramov, F Abuah, J Boike, JA Bradley, J Buongiorno, MA Cramm, RJ Giannone, D Giovanelli, Z Lyu, F Montemagno, S Mulay, *I Oduwole, R Paul, *JT Perez, S Peters, M Podar, I Shokodko, K Sipes, A Sukhanov, X Wu (2024). The complex system of organic carbon remineralization in rapidly-thawing Svalbard permafrost and active layer soils. DOE Users Meeting (Washington, DC).

Sipes, K, J Buongiorno, **AD Steen**, A Abramov, SL Peters, RJ Giannone, RL Hettich, J Boike, SL Garcia, TA Vishnevetskaya, KG Lloyd (2024). Bacterial MAGs diversity in the permafrost active layer of Ny-Ålesund, Svalbard. Geobiology Gordon Research Conference (Galveston, TX).

Abuah, F, K Sipes, J Buongiorno, AD Steen, JA Bradley, D Giovanelli, A Abramov, S Peters, RJ Giannone, RL Hettich, J Boike, SL Garcia, **AD Steen**, TA Vishnevetskaya, KG Lloyd. Subsurface microbes may drive climate feedbacks in thawing Arctic permafrost. Geobiology Gordon Research Conference (Galveston, TX).

Abuah, F, J Buongiorno, S Peters, K Sipes, JA Bradley, D Giovannelli, J Boike, A Abramov, T Vishnevetskaya, **AD Steen**, R Hettich, KG Lloyd (2024). Using multi-omics to link microbial organic

matter degradation in permafrost active layer soils to fjord sediments in Svalbard, Norway.

International Conference on Permafrost (Whitehorse, Canada).

- Steen, AD**, ^{*}I Oduwale, A Babjac, O Queen, KG Lloyd, SJ Emrich (2023). Is slow growth the major predictor of culturing difficulty for environmental microbes? AGU Fall Meeting (San Francisco, CA).
- ^{*}Perez, JT, ^{*}B Green, ^{*}M Stokes, MA Cramm, S Mulay, M Selci, K Sipes, A Abramov, JA Bradley, J Boike, D Giovannelli, TA Vishnevetskaya, KG Lloyd, D Keffer, **AD Steen** (2023). Molecular dynamics simulations of extracellular enzymes in active-layer soils from Svalbard. AGU Fall Meeting (San Francisco, CA)
- Mahmoudi, N, A Sanger, GP Halverson, KO Konhauser, **AD Steen** (2023). Earth before Exoenzymes: resolving the early evolution of microbial heterotrophy. Gordon Research Conference on Chemical Oceanography (Holderness, NH).
- Ruff, SE, IH de Angelis, M Mullis, JP Payet, BK Reese, KG Lloyd, **AD Steen**, HG Morrison, ML Sogin, The Census of Deep Life Consortium, J Ladau, F Colwell (2023). A global atlas of subsurface microbiomes reveals phylogenetic novelty, large scale biodiversity gradients, and a marine-terrestrial divide. 2nd Joint Symposium of the International Societies for Environmental Biogeochemistry & Subsurface Microbiology (Banff, Canada). DOI: 10.3897/aca.6.e108531
- Oduwale, I, TA Vishnivetskaya, K Sipes, AA Abramov, EM Rivkina, **KG Lloyd**, **AD Steen**. *In silico* investigation of the thermal adaptation of putatively thermophilic microbes in permafrost. Southeastern Biogeochemistry Symposium (Columbia, SC).
- Lloyd, KG, K. Sipes, J Buongiorno, M Tamino, **AD Steen**, A Abramov, S Peters, R Hettich, J Boike, SL Garcia, T Vishnivetskaya (2023). Keynote: Subsurface persistence of carbon degrading microbes as thawing permafrost moves between the terrestrial and marine environments. Goldschmidt Conference (Lyons, France).
- Abuah, C., K Sipes, MA Cramm, W Cyr, [†]B Green, M Essington, **AD Steen**, J Boike, JA Bradley, A Abramov, D Giovannelli, T Vishnivetskaya, KG Lloyd (2022). Investigating microbial communities in Svalbard Permafrost. AGU Fall Meeting (Chicago, IL).
- [†]Stokes, MH, R Paul, TJ Rogers, JM de Moor, A Stefánsson, SA Halldórsson, PH Barry, D Giovannelli, KG Lloyd, **AD Steen** (2022). Activities of extracellular enzymes in hot springs of southwestern Iceland. Southeastern Biogeochemistry Symposium (Atlanta, GA).
- [†]Mireles, C, ^{*}B Green, T Vishnivetskaya, KG Lloyd, **AD Steen** (2022). What are extracellular enzymes doing in permafrost? Exhibition of Undergraduate Research and Creativity (Knoxville, TN).
- Ruff, SE, JP Payet, IH de Angelis, KG Lloyd, CS Sheik, **AD Steen**, C Magnabosco, J Ladau, BK Reese, J Huber, A Shipunova, A Morozov, H Morrison, M Sogin, F Colwell (2022). High phylogenetic diversity and novelty in the global subsurface microbiome. International Sedimentological Congress (Beijing).
- ^{*}Oduwale, I, [†]M Hibbs, ^{*}TM Royalty, KG Lloyd, **AD Steen** (2021). Distribution of genomic novelty among Earth's environments in phylogenetically diverse uncultured microbes. American Geophysical Union Fall Meeting (New Orleans, LA).
- Buongiorno, J, EK Phillips, [†]M Stokes, C Coleman-King, **AD Steen** (2021). Outcomes and lessons learned from a partnership with East Tennessee Freedom Schools to involve K-12 students in authentic microbial ecology research. American Geophysical Union Fall Meeting (New Orleans, LA).
- [†]Stokes, MH, ^{*}B Green, A Abramov, T Vishnivetskaya, K Sipes, KG Lloyd, **AD Steen** (2021). Extracellular enzyme activities and mineralogy of thawing permafrost soils near Ny Ålesund, Svalbard. American Geophysical Union Fall Meeting (New Orleans, LA).
- ^{*}Royalty, TM, **AD Steen** (2021). Functional redundancy in ocean microbiomes controls trait stability. American Geophysical Union Fall Meeting (New Orleans, LA).
- Steen, AD**, ^{*}TM Royalty, ^{*}JM Schmidt, KG Lloyd (2021). Long enzyme lifetimes in subsurface sediments. American Geophysical Union Fall Meeting (New Orleans, LA).
- ^{*}Royalty, TM, **AD Steen** (2021). Trait Contribution Evenness—A functional redundancy metric for community-aggregated traits in microbial communities. U.S. Department of Energy Office of Biological and Environmental Research 2021 Biological System Sciences Division Principal Investigators' Meetings (Online)

- *Royalty, TM, **AD Steen** (2020). Trait Contribution Evenness – A functional redundancy metric for community-aggregated traits in microbial communities. American Geophysical Union Fall Meeting (online).
- Coleman-King, C, **AD Steen**, K Rearden, S Drumheller-Horton (2020). Making rocks matter: Utilizing critical and culturally responsive experiential learning in teacher education. International Conference on Urban Education (online).
- Lloyd, KG, K Sipes, T Vishnivetskaya, **AD Steen**, TC Onstott, R Liang, Z Garvin, RL Hettich, R Giannone, J Clift, J Bradley, J Boike, S Wetterich, A Abramov, E Rivkina (2020). Arctic Microbial Permafrost Degradation. DOE Genomic Sciences Program Annual Principal Investigator Meeting (Washington, DC).
- Sontag, P, K Dawson, [†]PJ Murray, **AD Steen** (2020). Influence of Biogeochemistry on Lead Availability in New Jersey Sediment: Development of Point-of-Use Trace Metal Sensor with Insights from Newark Biogeochemistry Teams. Ocean Sciences Meeting (San Diego, CA).
- Steen, AD**, C Burdysaw, SE Sawyer, *TM Royalty, KG Lloyd, RG Brook (2019). Diversity of functional genes in deeply branching uncultured microbes. AGU Fall Meeting (San Francisco, CA).
- *Royalty, TM, **AD Steen** (2019). Quantitatively partitioning microbial genomic traits among taxonomic ranks: implications for subsurface microbial communities. C-DEBI Networked Speaker Series (online). Archived at <https://www.darkenergybiosphere.org/outputs-resources/networked-speaker-series/nss-taylor-royalty/>.
- [†]Fang, C, *TM Royalty, **AD Steen** (2019). Analysis of positive selection on extracellular enzymes. UT-Knoxville EURēCA Undergraduate Research Symposium (Knoxville, TN)
- [†]Klibisz, A, *CS Cook, **AD Steen** (2019). The Potential of Oxidized Organic Sulfur as an Electron Acceptor for Organic Matter Respiration. UT-Knoxville EURēCA Undergraduate Research Symposium (Knoxville, TN)
- [†]Lonergan, A, K Fullerton, KG Lloyd, **AD Steen** (2019). Extracellular enzyme activity in hot springs in Panama. UT-Knoxville EURēCA Undergraduate Research Symposium (Knoxville, TN) and Southeastern Biogeochemistry Symposium (Columbia, SC)
- [†]Smith L, *TM Royalty, **AD Steen** (2019). Discerning the Relative Abundance of Extracellular DNA in Estuarine Sediments Using Spectrophotometry. UT-REU Microbial Community Interactions and Functions Poster Session (Knoxville, TN)
- [†]Miles, M, *TM Royalty, **AD Steen** (2019). The role of sulfonate in the sulfur and carbon cycles of estuarine sediments. UT-Knoxville EURēCA Undergraduate Research Symposium (Knoxville, TN)
- [†]Stooksbury, ZS, **AD Steen** (2019). Geological controls on biological carbon oxidation in the White Oak River Estuary, NC. UT-Knoxville EURēCA Undergraduate Research Symposium (Knoxville, TN) and Southeastern Biogeochemistry Symposium (Columbia, SC)
- [†]Ajuwon, F, [†]Malcolm X Shabazz Aquatic Geochemistry Team, [†]PJ Murray, KG Lloyd, **AD Steen** (2018). Breakdown of organic matter in freshwater: The role and nature of extracellular peptidases. AGU Fall Meeting (Washington, DC).
- [†]Mullen, LA, *TM Royalty, [†]Z Stooksbury, C Arnosti, **AD Steen** (2018). Activities of extracellular enzymes in sediments of the White Oak River estuary and Kongsfjord, Svalbard. AGU Fall Meeting (Washington, DC).
- *Royalty, TM, **AD Steen** (2018). Simulation-based approaches to characterize the effect of sequencing depth on the quantity and quality of metagenome-assembled genomes. AGU Fall Meeting (Washington, DC).
- Steen, AD**, *LA Mullen, *TM Royalty, C Arnosti (2018). Activities of heterotrophic microorganisms and their extracellular enzymes in Kongsfjorden, Svalbard. AGU Fall Meeting (Washington, DC).
- Steen, AD** (2018). Understanding the reactivity of proteinaceous OM as a property of microbial community metabolism. Gordon Research Conference on Organic Geochemistry (Plymouth, NH).
- Invited.**

- [†]Guidry, ME, ^{*}TM Royalty, **AD Steen** (2018). Protein Extraction from Sediment for Proteomic Analysis. UT-REU Microbial Community Interactions and Functions Poster Session (Knoxville, TN).
- Sontag, PT, [†]H Ofori, [†]Malcolm X Shabazz Aquatic Geochemistry Team, **AD Steen** (2018). Examining Influence of Raritan River-Bay Dissolved Organic Matter (DOM) and Extracellular Enzymes on Methylmercury (MeHg) in Phytoplankton. 10th Annual Sustainable Raritan Conference and Awards Ceremony (New Brunswick, NJ).
- [†]Ajuwon, FA, [†]PJ Murray, P Sontag, **AD Steen** (2018). Understanding the process of organic matter remineralization and its possible link to cycling of contaminants in estuaries. Southeast GSA (Knoxville, TN).
- ^{*}Mullen, LA, [†]Malcolm X Shabazz Aquatic Geochemistry Team, [†]K Boerrigter, [†]N Ferriero, J Rosalsky, [†]PJ Murray, **AD Steen** (2018). Potential activities of freshwater endo- and exo-acting extracellular peptidases in East Tennessee and the Pocono Mountains. Southeast GSA (Knoxville, TN).
- ^{*}Royalty, T, **AD Steen** (2018) Evaluating the sensitivity of subsurface microbial metagenome assembled genome properties as a function of metagenomic shotgun sequencing depth. Southeast GSA (Knoxville, TN).
- ^{*}Royalty, T, **AD Steen** (2018) Evaluating the sensitivity of subsurface microbial metagenome assembled genome properties as a function of metagenomic shotgun sequencing depth. Southeast Biogeochemistry Symposium. (Tallahassee, FL).
- [†]Stooksbury, Z, ^{*}LA Mullen, **AD Steen** (2018). Potential activities of extracellular glycosyl hydrolases, peptidases, alkaline phosphatase, and sulfatase in sediments of the White Oak River, NC. Southeast Biogeochemistry Symposium. (Tallahassee, FL).
- [†]Montes, D, [†]F Ajuwon, [†]E Uyoh, [†]W Aviles, [†]E Mends, [†]H Oyibo, [†]K Duarte, [†]D Akinawande, [†]S Nelson, [†]L Mozee, [†]E DeLima, [†]S Wellington, [†]N Ferriero, [†]PJ Murray, ^{*}L Mullen, PT Sontag, KG Lloyd, **AD Steen** (2018). Activities of extracellular enzymes in the Raritan Bay, NJ. SE GSA Meeting (Knoxville, TN).
- Buchan A, ^{*}LMN Quigley, BS O'Banion, RGM Spencer, A Stubbins, **AD Steen** (2018). Geo-metabolomics of a saltmarsh: Combining in situ, bulk, genomic, transcriptomic and DOMEomic data streams to understand microbial transformations of terrestrially derived organic matter. DOE Joint Genome Institute Annual Genomic of Energy & Environment User Meeting (San Francisco, CA).
- ^{*}Royalty, T, **AD Steen** (2018). Evaluating the sensitivity of subsurface microbial metagenome-assembled genome properties as a function of metagenomic shotgun sequencing depth. SE GSA meeting (Knoxville, TN).
- ^{*}Royalty, T, **AD Steen** (2017). A Sensitivity Analysis on Metagenome-Assembled Genome Response to Sequencing Depth: How Much Sequencing Is Enough? C-DEBI Annual Meeting (Marina, CA).
- Drumheller, SK, CD Sumrall, TA Gates, **AD Steen**, SP Horn, LD McKay (2017). Including citizen and student scientists in actualistic taphonomic research: The good, the bad, and the smelly. GSA Annual Meeting (Seattle, WA).
- ^{*}Quigley, LNM, **AD Steen**, A Edwards, A Buchan. Evidence for species-specific degradation of recalcitrant organic matter in the presence of labile organic compounds. ASM Microbe 2017 (New Orleans, LA).
- [†]Souder, RA, ^{*}LA Mullen, **AD Steen** (2017). Hydroxyl radicals may help heterotrophic microbes mineralize recalcitrant organic matter. Southeastern Biogeochemistry Symposium (Athens, GA).
- Steen, AD**, ^{*}LNM Quigley, A Stubbins, RG Spencer, A Buchan (2017). Towards a more mechanistic understanding of the aquatic priming effect. ASLO Aquatic Sciences Meeting (Honolulu, HI).
- ^{*}Barrett, AvB, [†]Malcolm X Shabazz Aquatic Geochemistry Team, [†]PJ Murray, **AD Steen** (2017). Environmental controls on protein degradation pathways in freshwaters of eastern Tennessee and northeastern Pennsylvania. ASLO Aquatic Sciences Meeting (Honolulu, HI).
- [†]Boerrigter, KA, **AD Steen**, J Rosalsky, [†]N Ferriero, [†]PJ Murray (2016). Study of Various Peptidases and Their Effects on Freshwater Bodies. AGU Fall Meeting (San Francisco, CA).

- Mahmoudi, N, SR Beaupre, **AD Steen**, A Pearson (2016). Investigating the bioavailability and degradation of sedimentary organic matter in Guaymas Basin using the IsoCaRB system. C-DEBI Annual Meeting (Marina, CA).
- Mahmoudi, N, SR Beaupre, **AD Steen**, A Pearson (2016). Sequential bioavailability of sedimentary organic matter to heterotrophic bacteria. Gordon Research Conference (GRC) Organic Geochemistry (Holderness, NH).
- Mahmoudi, N, SR Beaupre, **AD Steen**, A Pearson (2016). Investigating microbial C cycling in sediments using the IsoCaRB system. American Society for Microbiology (ASM) Microbe Meeting (Boston, MA).
- McKay, L.D., SP Horn, KA Affholter, C Cropper, SK Drumheller-Horton, KN Ellis, K Fristoe, J Jones, CS Hadjiharalambous, AC Lee, **AD Steen**, CD Sumrall (2016). Engaging community college transfer students in Tennessee to improve recruitment, retention and success in 4-year geoscience programs. Geological Society of America Annual Meeting (Denver, CO).
- Steen, AD**, [†]JP Vazin, [†]KH Mulligan, [†]SM Hagen, SW Wilhelm (2016). What we talk about when we talk about peptidases: Quantifying the substrate specificities of aminopeptidases in fresh and estuarine water. 5th International Enzymes in the Environment Conference. (Bangor, Wales)
- Drumheller, SK, TA Gates, [†]LR Brett, **AD Steen** (2016). Root marks in the classroom: A student science approach to actualistic taphonomic research. Society of Vertebrate Paleontology Annual Meeting (Salt Lake City, UT)
- Mahmoudi N, S Beaupre, **AD Steen**, A Pearson (2016). IsoCaRB: A novel bioreactor system to characterize microbial carbon cycling and the lability of natural organic matter. 16th International Symposium on Microbial Ecology (Montreal)
- ^{*}Schmidt, JM, **AD Steen** (2016). Potential activities of organic carbon degrading extracellular enzymes in Baltic Sea sediments. Southeastern Biogeochemistry Symposium (Knoxville, TN)
- Steen, AD**, ^{*}JM Schmidt (2016). The nature and function of microbial enzymes in subsurface marine sediments. Ocean Sciences Meeting (New Orleans, LA)
- [†]Murray, PJ, [†]N Ferriero, J Rosalsky, KG Lloyd and **AD Steen** (2016). How do high school students respond to opportunities to collaborate with authentic scientific researchers in at-risk environments? Ocean Sciences Meeting (New Orleans, LA)
- ^{*}Schmidt, JM, **AD Steen** (2015). Potential activities of extracellular enzymes in the deep subsurface sediments of the Baltic Sea basin. GSA Annual Meeting. Published in GSA Abstracts with Programs, 47 (7).
- Steen, AD** (2015). How to use extracellular enzyme biochemistry to better predict the effect of increasing CO₂ on microbe-organic matter interactions. Goldschmidt 2015 (Prague).
- Steen, AD**, [†]PJ Murray, [†]N Ferriero, [†]Malcolm X Shabazz Aquatic Geochemistry Team, J Rosalsky, KG Lloyd (2015). Activities of extracellular peptidases across environments: Do diverse aquatic environments share similar tastes? ASLO Aquatic Sciences Meeting, Granada, Spain.
- Steen, AD**, [†]PJ Murray, [†]N Ferriero, [†]Malcolm X Shabazz Aquatic Geochemistry Team, J Rosalsky, KG Lloyd (2015). Involving at-risk high school student populations in environmental research: A case study using extracellular enzyme assays. ASLO Aquatic Sciences Meeting, Granada, Spain.
- Lloyd, KG, J Bird, **AD Steen**, J Buongiorno, S Jungbluth, IODP Leg 347 Shipboard Science Party (2015). Insights from the first deep subsurface single cell genomes and development of digestR, a new tool for metagenomic analysis. C-DEBI NSF Site Visit, Los Angeles, CA.
- Steen, AD**, ^{*}L Mach (now LNM Quigley), A Buchan (2014). The marine priming effect: Stimulation of detrital organic matter remineralization by addition of labile organic carbon in an estuarine microbial community. Joint Aquatic Sciences Meeting, Portland, OR.
- Steen, AD**, K Michalska, G Chhor, M Endres, [†]JP Vazin, KG Lloyd, SW Wilhelm, A Joachimiak (2014). Strategies to assess the biochemical properties of extracellular hydrolases in aquatic environments. Goldschmidt Meeting, Sacramento CA.

- Michalska, K, **AD Steen**, G Chhor, K Fayman, M Endres, G Babnigg, KG Lloyd, R Jedrzejczak, A Joachimiak (2014). Towards understanding microbial life in marine sediments – Structure and specificity of a novel aminopeptidase. 23rd Congress of the International Union of Crystallography. Montreal, Canada.
- Arnosti, C, L D'Ambrosio, **AD Steen**, A Teske (2014). Site- and depth-related contrasts in enzymatic capabilities of microbial communities: Exploring links between microbial community function and composition. ASLO Ocean Sciences Meeting, Hawaii.
- Steen, AD**, RT Kevorkian, E Deas, KG Lloyd (2013). Activities of extracellular peptidases in sediments of the White Oak River Estuary, NC. C-DEBI Annual Meeting, Monterrey, CA.
- Lloyd, KG, M May, R Kevorkian, **AD Steen** (2013). A meta-analysis reveals methodological biases in quantitative methods for marine subsurface microbiology. Goldschmidt Conference, Florence, Italy.
- Arnosti, C, Z Cardman, **AD Steen**, K Ziervogel, A Teske (2013). Functional contrasts and functional redundancy in Arctic bacterial communities in the oxic water column and anoxic sediments. Goldschmidt Conference, Florence, Italy.
- [†]Vazin, JP, **AD Steen**, SW Wilhelm (2013). What's for Dinner? The Study of Substrate Specificities of Extracellular Peptidases in the Tennessee River to Determine Microbe Consumption Options. Exhibition of Undergraduate Research and Creative Achievement. Knoxville, TN. ***Winner of UT "EURECA" Undergraduate Research Award.**
- [†]Webber, AW, A Buchan, **AD Steen**, SW Wilhelm (2013). Effects of viral lysis on carbon cycling in marine microbial communities: tracking size fractions of released dissolved organic carbon. Exhibition of Undergraduate Research and Creative Achievement. Knoxville, TN. ***Winner of UT "EURECA" Undergraduate Research Award.**
- Steen, AD**, [†]JP Vazin, [†]AW Webber, PJ Gainer, SW Wilhelm (2013). Leucyl aminopeptidase is not enough: what do extracellular peptidase activity measurements tell us? ASLO Aquatic Sciences meeting, New Orleans, LA.
- Steen, AD**, R Cory, P Lee, DM McKnight, J Mikucki (2012). Fluorescence spectroscopy and biogeochemical analyses suggest chemosynthetic supply of reactive DOM to Lake Bonney. AGU Fall Meeting, San Francisco, CA.
- [†]Webber, AP, [†]JP Vazin, A Buchan, **AD Steen**, SW Wilhelm (2012). Effects of viral lysis on carbon cycling in marine communities: Assaying extracellular enzymes and tracking size fractions of released dissolved organic carbon. ASM KY-TN Branch meeting, Maryville, TN.
- Steen, AD**, C Arnosti (2011). Peptidase activity in an Arctic fjord: high aminopeptidase activity in cold water. Enzymes in the Environment workshop, Bad Nauheim, Germany, July 2011 ***Winner of Outstanding Poster award.**
- Ziervogel, **AD Steen**, M Piehler. (2011). Formation and abundance of organic gel-like particles in Bogue Sound, NC. 11th International Estuarine Biogeochemistry Symposium, Morehead City, NC.
- C Arnosti, **AD Steen**, K Ziervogel, S Ghobrial, WH Jeffrey (2010). Latitudinal gradients in degradation of marine dissolved organic carbon. ASLO Aquatic Sciences Meeting, Santa Fe, NM.
- Ziervogel, K, **AD Steen**, S Ghobrial, C Arnosti (2010). Advective flow over permeable sandy sediments enhances extracellular enzyme activities in overlying waters. ASLO Ocean Sciences meeting, Portland, OR.
- Steen, AD**, C Arnosti (2009). Extracellular enzymatic activity in the water column: A comparison among polar, temperate, and subtropical environments. International Meeting on Organic Geochemistry, Bremen, Germany.
- Arnosti, C, **AD Steen**, K Ziervogel, S Ghobrial, WH Jeffrey (2009). Latitudinal gradients in microbial enzyme activities: Implications for DOC cycling in changing oceans. Gordon Research Conference in Chemical Oceanography, Tilton, NH.

- Steen, AD**, C Arnosti (2009). Extracellular enzymatic activity in the water column: A comparison among polar, temperate, and subtropical environments. Chemical Oceanography in a Changing World, Savannah, GA. ***Winner of Best Student Poster award.**
- Ziervogel, K, **AD Steen**, C Arnosti (2009). Enhanced extracellular enzyme activity in marine snow and in the surrounding water: Consequences for carbon cycling in coastal waters. Chemical Oceanography in a Changing World, Savannah, GA.
- Steen, AD**, K Ziervogel, S Ghobrial, C Arnosti (2009). Microbial turnover of high molecular weight dissolved organic matter: Where are the roadblocks? ASLO Aquatic Sciences, Nice, France.
- Robador, A, V Brüchert, **AD Steen**, and C Arnosti (2009). Temperature sensitivity of extracellular enzymatic hydrolysis in Arctic and temperate marine sediments. ASLO Aquatic Sciences, Nice, France.
- Steen, AD**, C Arnosti (2008). Degradation rates of extracellular enzymes in polar and subtropical seawater: Implications for bioavailability of high molecular weight organic carbon. ASLO Ocean Sciences, Orlando, FL. ***Winner of Outstanding Student Talk award.**
- Robador, A, V Brüchert, **AD Steen**, C Arnosti (2008). Temperature sensitivity of bacterial carbohydrate hydrolysis in Arctic and temperate marine sediments. Aquashift workshop, Konstanz, Germany.
- Arnosti, C, L Hamdan, **AD Steen** (2007) Contrasting dynamics of high molecular weight dissolved organic carbon in the Chesapeake Bay and coastal ocean: Insights from enzyme activities, carbohydrate inventories, and microbial metabolism. Ocean Carbon and Biogeochemistry Summer Workshop. Woods Hole, MA.
- Steen, AD**, L Hamdan, C Arnosti (2007). PowerBars vs. Cardboard: Small structural differences determine the fate of polysaccharides in an estuary. ASLO Aquatic Sciences, Santa Fe, NM.
- Steen, AD**, C Arnosti (2005). The influence of marine polymer gels on the activity of extracellular enzymes. ASLO Ocean Sciences, Salt Lake City, UT.
- Steen, AD**, C Arnosti, NV Blough (2004). Electron paramagnetic resonance spectroscopy used to measure polysaccharide hydrolysis and macromolecule-surface interactions in aquatic environments. Gordon Conference in Organic Geochemistry, Holderness, NH.
- Steen, AD**, C Arnosti, NV Blough (2003). A novel approach to measuring macromolecule-surface interactions and extracellular enzyme activities in sediments. Hedges Symposium: New Approaches in Marine Organic Biogeochemistry; Friday Harbor, WA.

Conference sessions proposed or co-chaired

- Co-proposer and co-chair (2023). Microbial and Abiotic Factors Influencing the Turnover and Fate of Organic Matter in Aquatic Systems. ASLO Aquatic Sciences Meeting, Palma de Mallorca, Spain.
- Co-chair (2021). Geomicrobiology of the Deep Biosphere. AGU Fall Meeting, New Orleans, LA.
- Co-proposer and co-chair (2021). Innovative Approaches in Geoscience Education via University-K12 Partnerships I Oral. AGU Fall Meeting, New Orleans, LA.
- Co-proposer and co-chair (2018). CT14B Tracing Ocean Processes: From Organic Carbon Oxidation to Age Estimates by Transient Tracers. ASLO/AGU/TOSS Ocean Sciences Meeting. Portland, OR.
- Co-proposer and co-chair (2016). Microbial Enzymes in Aquatic Environments: Environmental Control, Genomics, and Biogeochemical Functions. ASLO/AGU/TOS Ocean Sciences Meeting. New Orleans, LA.

Lectures for the general public

- Steen, AD** (2022). Soil Microbiology in Svalbard, or, How we learn about the littlest environmental engineers.
- Steen, AD** (2016). Heterotrophic organisms in deep marine sediments, or, What to do when you're hungry but you're 100,000 years late for dinner. ORION (Oak Ridge amateur astronomy and science club) monthly meeting (Oak Ridge TN).

Participation in workshops

These represent workshops or other small meetings that required applications or invitations to attend.

2022	Mapping the future of 'omics applications in Antarctic Research. Virtual.
2022	International Continental Drilling Project workshop: Deep drilling in the Turkana Basin: Exploring the link between environmental factors and hominin evolution
2019	Hanse-Wissenschaftskolleg Workshop on Marine Organic Biogeochemistry, Delmenhorst, Germany.
2018	Gordon Research Conference on Organic Geochemistry, Holderness, NH.
2016	Center for Dark Energy Biosphere Research Annual Meeting, Marina, CA, USA.
2015	Deep Carbon Observatory Early Career Scientist Workshop, Ponta Delgada, Azores, Portugal.
2014-2021	Southeastern Biogeochemistry Symposium (annual), various locations, USA.
2013	Center for Dark Energy Biosphere Research Annual Meeting, Marina, CA, USA.
2011	Emerging Issues Seminar: Microbial Carbon Pump in the Ocean, San Juan, Puerto Rico, USA.
2009	Enzymes in the Environment Workshop, Bad Neuheim, Germany.
2009	Chemical Oceanography in a Changing World; Savannah, GA.
2005	Gordon Research Conference on Organic Geochemistry; Holderness, NH.
2003	Symposium on Cold-Adapted Microbes; Bremen, Germany.
2003	Hedges Symposium: New Approaches in Marine Organic Biogeochemistry; Friday Harbor, WA, USA.

Honors and awards

2013 & 2014	Excellence in Reviewing award, Biogeochemistry (top 25 reviewers)
2013	Trainee on UNOLS Chief Scientist Training Cruise, \$1500
2011	Outstanding Poster Award (researchers under age 35), Enzymes in the Environment: Activity, Ecology, and Applications Workshop, Bad Nauheim, Germany
2009	Best Student Poster Award, Symposium on Chemical Oceanography in a Changing World, Savannah GA
2008	Outstanding Student Talk award, Ocean Sciences meeting, Orlando, FL
2013	ASLO Early Career Travel Grant, \$500
2011	FEMS Young Scientist Meeting Grant, \$550
2006	EPA Star Fellowship (3 years PhD Fellowship including full stipend and tuition), value approx. \$160,000
2004	Agouron Institute scholarship to attend USC Wrigley Institute Geobiology summer course, value approx. \$4000

Student & postdoc mentorship

Major Graduate Student and Postdoc Advisees:

2024-present	Eesha Rangani (PhD, Microbiology).
2023-present	Meghan Winzler (PhD, Microbiology): Physiology of subsurface heterotrophs.
2022-present	Kambiz Kalhor (PhD, Microbiology): Machine learning approaches to understand subsurface microbial ecology.
2022-2024	Adrianna Matthews (no degree, Microbiology). Metabolism of east Antarctic shelf sediment heterotrophs.
2021-present	Jacob Perez (PhD, Geology): Biogeochemical consequences of enzyme-clay interactions in thawing permafrost.

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2021-present	Iyanu Oduwole (PhD, Genome Science and Technology): Analysis of genes for secreted enzymes in uncultured microbes.
2020-2022	Brianna Green (no degree, Geology): Microbe-organic matter interactions in thawing permafrost
2020-2021	Taylor Royalty (Postdoctoral associate): Functional redundancy in microbial ecosystems
2017-2020	Taylor Royalty (PhD, Geology): Theoretical and empirical methods relating metabolism diversity with microbial phylogenetics, community diversity, and organic geochemistry
2018-2020	Christopher S. Cook (MS, Geology): Effects of enzyme assay protocol on microcystin and other organic matter degradation studies
2018-2020	Lauren A. Mullen (MS, Geology): Effect of temperature on heterotrophic processes: A case study comparing temperate and Arctic sediments
2018-2019	Lauren Krausfeldt (Postdoctoral associate; with Steven Wilhelm, UT-K Microbiology)
2015-2017	Abigail V. Barrett (MS, Geology): Controls on microbial extracellular enzymes in northeast Pennsylvania and eastern Tennessee fresh waters
2014-2016	Jenna M. Schmidt (MS, Geology): Microbial extracellular enzymes in marine sediments: Methods development and potential activities in the Baltic Sea deep biosphere.

Graduate student committee memberships (UTK):

Ashley Berg (Ph.D., Earth and Planetary Sciences), Kathleen Brannen (Ph.D., EPS), Walt Doty IV (MS, EPS), Chanda Drennen (Ph.D., EPS), Aaron Goemann (MS, EPS), Miles Henderson (Ph.D., EPS), Jordan Bird (Ph.D., Microbiology), Lauren Mach Quigley (Ph.D., Microbiology), Charles Paradis (Ph.D., EPS), Jessica Stevens (Ph.D., Biosystems Engineering and Soil Science), Joshua Stough (Microbiology), Andrew Putt (Ph.D., EPS), Samantha Gwizd (Ph.D., EPS), Helena Pound (Ph.D., Microbiology), Isis Fukai (Qualifying Exam committee, Ph.D., Bredesen Center), Grace Sarabia (Ph.D., Chemistry), Eric Fussell (CRP, Ph.D., Chemistry), Caleb Garrett (CRP, Ph.D., Chemistry), Diana Ramirez (Ph.D., Microbiology), Jill Walton (Ph.D., Microbiology), Allison Rogers (MS, EPS), Jennifer Bailey (Ph.D., Microbiology), Sayli Mulay (Ph.D., Microbiology), Ethan Sweet (Ph.D., Earth and Planetary Sciences), Margaret Spangler (Qualifying Exam committee, Ph.D., Genome Science and Technology), Mujeeb Wakeel (M.S., Chemistry), Katelyn Houghton (Ph.D., Microbiology)

Undergraduate advisees (UTK):

UTK: Logan Brett, Andrew Crowley, Shane Hagan, Drew Mather, Lauren Mullen, Katherine Mulligan, Rebecca Pinals, Gabrielle Rimmer, Shannon Ryan, Austen Webber, Jasmine Vazin, Josh Gurka, Daniel Shyles, Collin McCrilis, Aaron Souder, Patrice Dansby, Rebecca Craft, Zachery Stooksbury, Andrew Lonergan, Maryn Miles, Kaelyn Gurka, Adam Klibisz, Cameron Fang, James Dallas Dunn, Murray Stokes, Mattie Hibbs, Rily Majors, Cheyenne Mireles, Jerry McCrary, Nolan Wells.

USC: Gabriel Velazquez, Annnika Harusadangkul, Daryl Christi (as non-thesis MS student); Arib Ahsan, Daniel Pirich, Sasha Gupta, Max Khackikian (UC-Davis undergraduate student); .

Teaching

Classes developed or totally overhauled:

- FYS 129: Mountains and Oceans
- GEOL 456 / 558: Global Climate Change
- GEOL 301: Methods in Geoscience
- GEOL 660: Seminar in Geochemistry: Organic Matter on the Earth and Elsewhere
- MICR 475 / 575: Reproducible Data Analysis

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Classes taught:

- FYS 129: Mountains and Oceans (x1)
- GEOL 103: The Earth's Environments (x3)
- GEOL 301: Field Methods in Geology (x1)
- GEOL 456 / 558: Global Climate Change (x4)
- GEOL 660: Seminar in Geochemistry (x1)
- MICR 475 / 575: Reproducible Data Analysis (x4)

Short courses and independent studies taught:

Fall 2017 2-day course: Software Carpentry
Spring 2016 Independent study: "Introduction to organic geochemistry"
Spring 2014 Independent study: "Aquatic extracellular enzymes: Theory and practice"
Spring 2011 1-day course: "Basics of the R Statistical Platform"

Post-baccalaureate laboratory technicians supervised:

2014-2015 Austen Webber. Now employed as a microbiologist at a craft brewing company.
2015-2018 Lauren Mullen. Joined the lab as a Masters' student after working as a technician, graduated Spring 2020. Now employed as an environmental microbiologist in industry.

Leadership Positions

2021-present Founder of East Tennessee Freedom School Aquatic Ecology Team, an initiative to involve middle school and high school students in authentic environmental science research.

2018-2019 Director of Environmental Studies Concentration.

2014-2018 Assistant Director of Environmental Studies Concentration.

2018 Co-organized and co-taught [Software Carpentry](#) workshop (2-day workshop for scientists to learn better coding & data analysis skills, taken by 36 students from all around the University)

2017-2019 Collaboration with Girl Scouts of the Southern Appalachians to develop climate science and geomicrobiology-related activities for Cadette-level Girl Scouts.

2016-2017 Guest associate editor, *Frontiers in Microbiology* Research Topic: Extracellular enzymes in aquatic environments: exploring the link between genomic potential and biogeochemical consequences. Published as an ebook at <https://www.frontiersin.org/research-topics/4814/extracellular-enzymes-in-aquatic-environments-exploring-the-link-between-genomic-potential-and-bioge>

2012-present Co-founder of the Malcom X Shabazz High School Aquatic Biogeochemistry Team in Newark, NJ to study carbon degradation pathways in water bodies near Newark. To date ~70 students have participated in this program. These data have formed the core of one paper (Mullen et al, 2018) and four presentations at international (AGU) and regional meetings.

Service to the Scientific Community

2019 Co-organizer of C-DEBI workshop, "Microorganisms and Organic Carbon in the Marine Subsurface".

2015 Blogger for 2015 Goldschmidt meeting. Posts were published as part of the European Association of Geochemistry's blog, <http://blog.eag.eu.com/>.

2009-present Reviewer for funding agencies including the National Science Foundation (USA), NASA, DfG (German national science foundation), FWF (Austrian Science Fund), FONDECYT

(Chilean national science foundation), *Applied and Environmental Microbiology*, *Aquatic Geochemistry*, *Aquatic Microbial Ecology*, *Biogeosciences*, *Biogeochemistry*, *Environmental Geotechnics*, *Environmental Microbiology*, *FEMS Microbiology Ecology*, *Frontiers in Marine Science*, *Frontiers in Microbiology*, *Geochimica et Cosmochimica Acta*, *Geoderma*, *ISME Journal*, *Limnology and Oceanography*, *Marine Chemistry*, *Marine Drugs*, *Marine Environmental Research*, *PLOS ONE*.

2011-present Poster judge at meetings of the Association for the Sciences of Limnology and Oceanography.

2005-2006 President, UNC Marine Science Graduate Action Group

Professional society memberships

- American Society for Microbiology
- American Geophysical Union
- American Chemical Society
- Black Microbiologists Association (Affiliate member)

Field experience

2024 Chief Scientist, R/V Sikuliaq, cruise SKQ-2024-10S, 15 days.

2022 Participant: MARSSAM sediment coring PI training cruise, R/V Roger Revelle, 10 days.

2022 Expedition leader: 2-week expedition to Svalbard for permafrost sampling.

2018 Expedition participant: MPI-Bremen / Center for Geomicrobiology 2-week expedition to Svalbard for fjord sediment sampling.

2012 Cruise participant: NSF/UNOLS Chief Scientist Training Program cruise, R/V Hugh R. Sharp. 5-day cruise plus 2-day classroom component.

2012 Cruise participant: R/V Savannah, 8 days, offshore North and South Carolina, USA.

2012-present Led sampling at the White Oak River estuary, NC, USA, 1-3 times per year.

2010 Expedition organizer: MPI-Bremen / Center for Geomicrobiology expedition to Svalbard. 3 weeks at sea and at AWIPEV base, Ny Ålesund.

2005 Course participant: Agouron International Geobiology Course (6 week summer course, including one-week field camp in the Canadian Rockies).

2004-2008 Expedition participant: Four, 2 to 4-week expeditions to Svalbard, including 2-8 day expeditions on M/V Farm and sample processing/experimental work at AWIPEV base, Ny Ålesund.

2003 Cruise participant: R/V Cape Henlopen, 5 days, Chesapeake Bay and offshore Virginia, USA.